# TD02_Confidence_intervals

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## 1 TD 2 : Confidence interval

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### 1.1 1. Data presentation

**Session objective**

**Based on a proposed data set and the biological problem raised, the aim is to describe the sample, an essential step prior to any statistical analysis and make your first steps in the statistical programming language R.**

A clinician evaluates the intensity of depression on the Hamilton scale, which ranges from 0 (not depressed) to 56 (very depressed), in 100 patients suffering from a Major Depressive Episode (MDE). We want to know whether acupuncture improves the course of depression. To this end, 100 patients with the same depressive state (same Hamilton score) were divided into 2 groups of 50 by drawing lots. by lot. Group I received pharmacological treatment only, while Group II received pharmacological treatment combined with acupuncture. pharmacological treatment combined with acupuncture. The dataset contains the following variables: - `hamilton_g1`: Hamilton scores, after 8 months of treatment, for patients of group I; - `hamilton_g2`: Hamilton scores, after 8 months of treatment, for patients of group II; - `urbanization_g1`: level of urbanization in which patients live (A: rural, B: moderate urbanization, C: highly urbanized) for patients of group I; - `urbanization_g2`: level of urbanization in which patients live (A: rural, B: moderate urbanization, C: highly urbanized) for patients of group II; - `age_g1`: patients' age in years for group I; - `age_g2`: patients' age in years for group II; - `weight_g1`: patients' weight in kilograms for group I; - `weight_g2`: patients' weight in kilograms for group II; - `sex_g1`: patients' gender in group I; - `sex_g2`: patients' gender in group II.

The data is contained in the text file `data_depression.txt` in the `data/` directory, whose full path is `data/data_depression.txt`.

Data are read using R's `read.table()` function:

[2]:
```r
# Read data file
depression = read.table("data/data_depression.txt", header = TRUE)

# Display start of data file contents
head(depression)
```

A data.frame: 6 × 10

| | hamilton_g1 <int> | hamilton_g2 <int> | urbanization_g1 <chr> | urbanization_g2 <chr> | age_g1 <int> | age_g2 <int> |
|---|---|---|---|---|---|---|
| 1 | 43 | 25 | B | B | 57 | 48 |
| 2 | 29 | 29 | A | B | 37 | 41 |
| 3 | 34 | 18 | A | A | 7 | 30 |
| 4 | 39 | 23 | A | A | 37 | 41 |
| 5 | 40 | 24 | B | A | 15 | 30 |
| 6 | 41 | 25 | B | B | 45 | 45 |

## 1.2 2. Confidence Interval

**Why calculate a confidence interval?**

The aim of estimation is to determine the unknown values of the population parameters (proportion, mean, variance) from sample data. It is a function of the sample observations, whose value will be used in place of the true the true value of the parameter. The value taken by the parameter on a given sample is called an estimate. The **estimator** is a **random variable** with a **mean** and a **variance**. and a **variance**.

***When moving from one sample to another, the estimator remains the same (for example the mean) but its value varies, as shown in the example below on the two samples of 100 vaccines, there are two estimates of the frequency of non-compliant vaccines: 5% for group A and 3% for group B.***

Example:

Estimated frequency in two different samples A and B.

The value of the estimate fluctuates according to the sampling: point estimation methods only concern data from a sample, but in a population, it is possible to choose different samples.

It is therefore impossible to have the true mean value in the population studied. What's more, estimating a mean from a sample is of little interest if you have no idea of the precision of the estimate obtained. This is why we look for

## 1.3 3. Application to an example of confidence interval calculation

**Question: What is the confidence interval of the mean Hamilton score of group 1 for female individuals?**

### 1.3.1 3.1 Building the dataset to answer the question

To answer this question, we first need to : - construct a dataset containing all the Hamilton scores of group 1 for female individuals. - display the sample size - plot histogram - plot boxplot

[3]:
```
# Data extraction
women_g1 = depression[depression$sex_g1 == "F", "hamilton_g1"]

# Display extracted data
women_g1
```

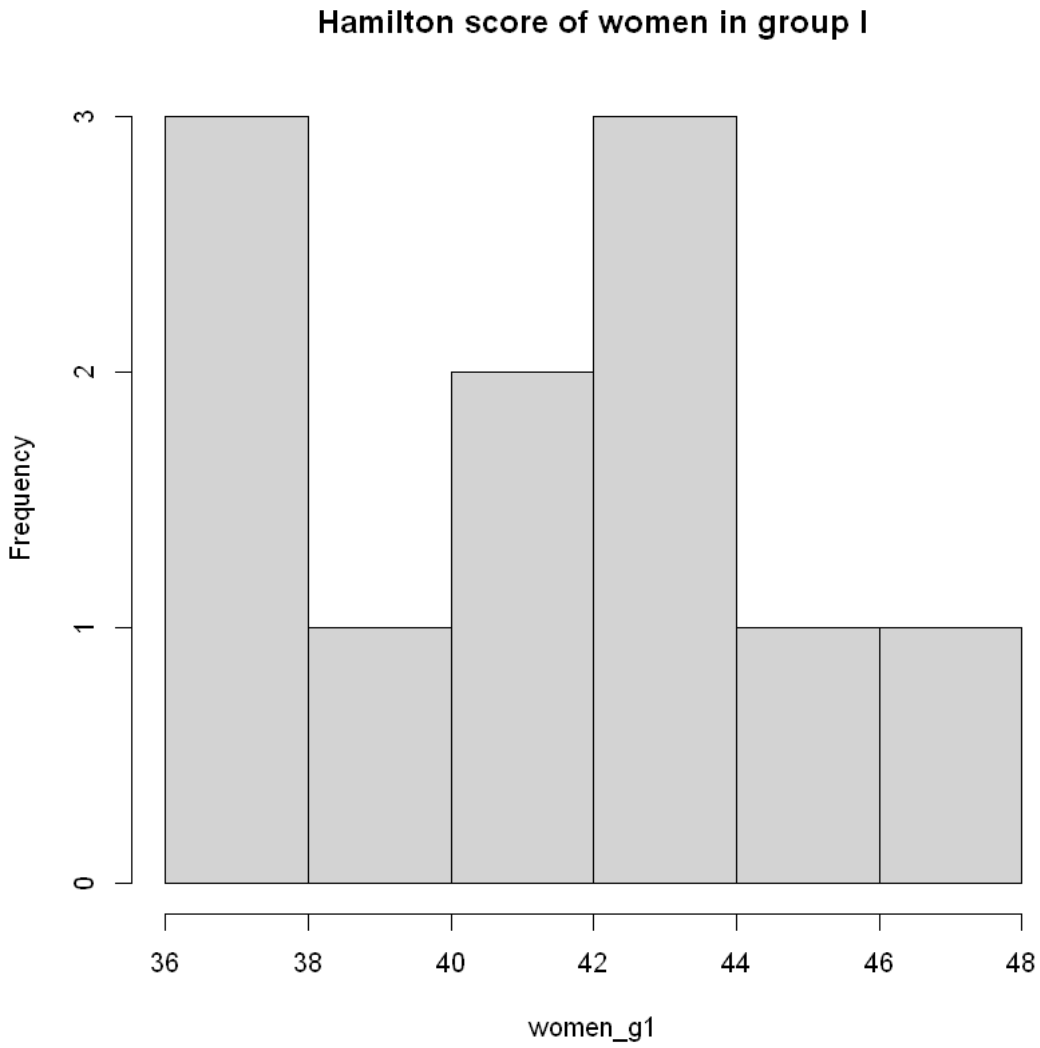
```r
# Sample size
length(women_g1)

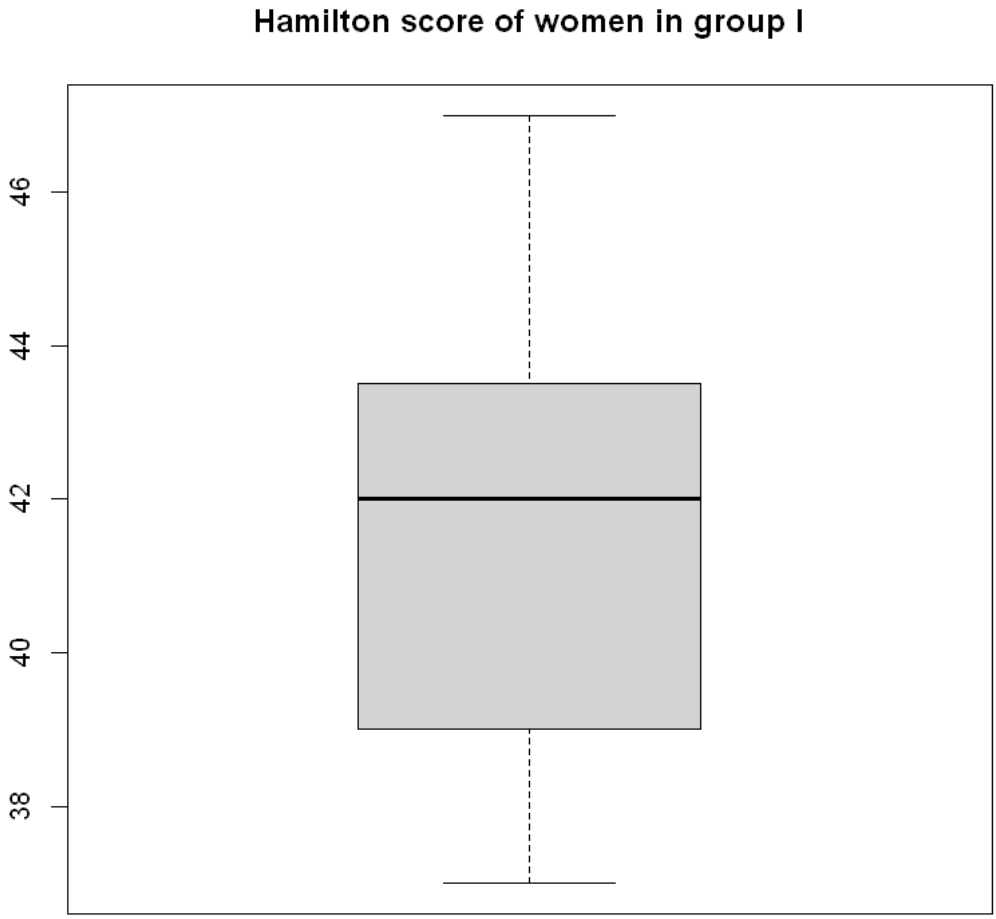
# Plot histogram
hist(women_g1, main = "Hamilton score of women in group I")

# Plot boxplot
boxplot(women_g1, main = "Hamilton score of women in group I")
```

1. 43 2. 37 3. 38 4. 40 5. 37 6. 41 7. 43 8. 47 9. 46 10. 42 11. 44

11






### 1.3.2 3.2 Calculating sample parameters

Describe the sample by calculating : - mean - variance - standard deviation - display statistical summary

Comment on your results

[4]:
```
# Average
mean(women_g1)

# Variance
var(women_g1)
```

```
# Standard deviation
sd(women_g1)

# Statistical summary
summary(women_g1)
```

41.6363636363636

11.6545454545455

3.41387543043759

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  37.00   39.00   42.00   41.64   43.50   47.00
```

**Comment:**

Given what we know of the rest of group I, it seems that women have a higher Hamilton score than men. Also we have n = 11 which is lower than 30 so we need to check that the score follow a normal distribution. According to our graph it doesn't seem so.

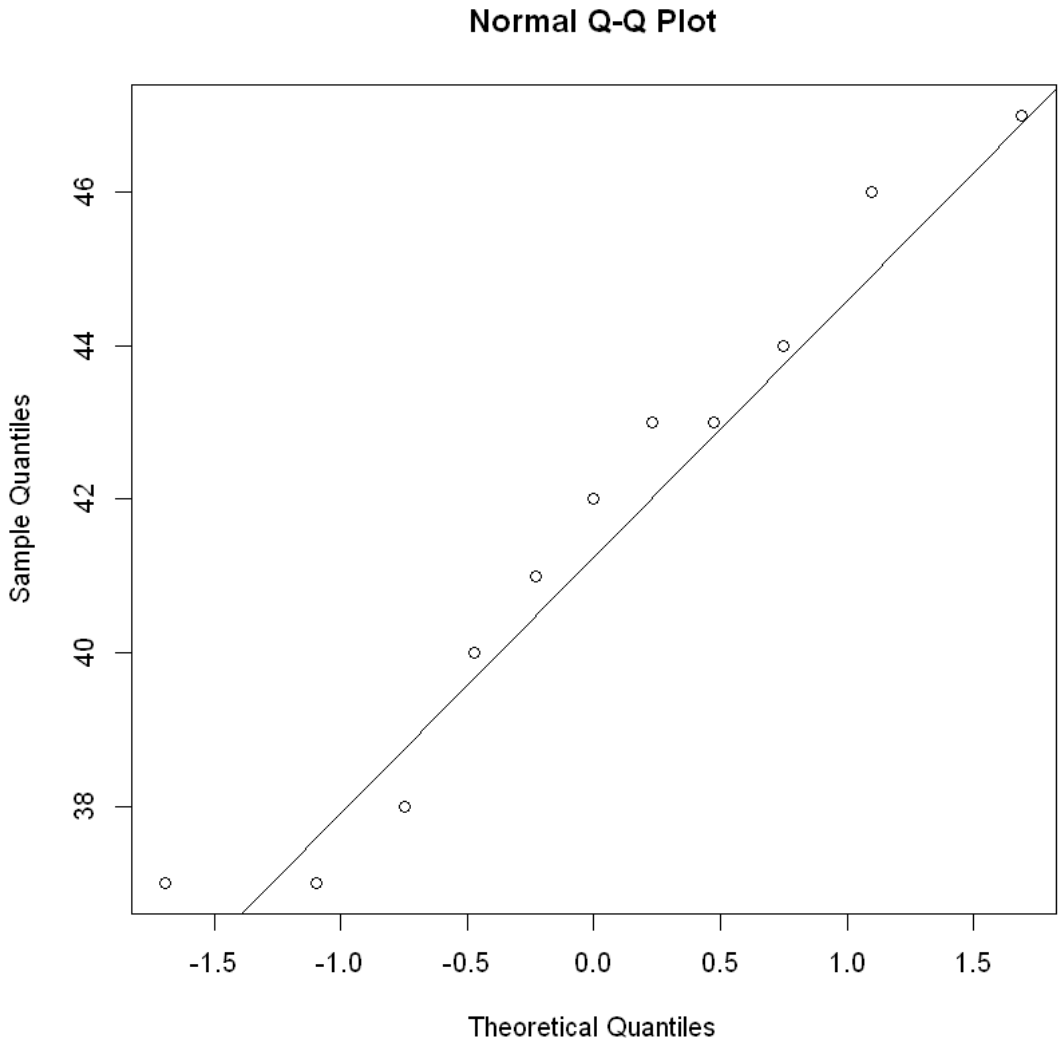
**Before calculating the confidence interval, you need to check the conditions of application:** If the sample size is large ($n \geq$): no conditions of application. - If the sample size is large ($n \geq 30$): no application condition - If the sample size is small ($n < 30$): the data must follow a normal distribution.

### 1.3.3 3.3 Sample normality test

**Graphical method: the quantile-quantile graph** A graphical method is used to verify normality by plotting the quantiles of the sample against the quantiles of a normal distribution. quantiles of a normal distribution. Under the assumption of normality the points must line up on a straight line.

You can use the functions `qqnorm()` then `qqline()` on the data you want to display this quantile-quantile graph.

[5]:
```
# Graphical representation of quantiles of observations and quantiles of the␣
↪normal distribution
qqnorm(women_g1)
qqline(women_g1)
```



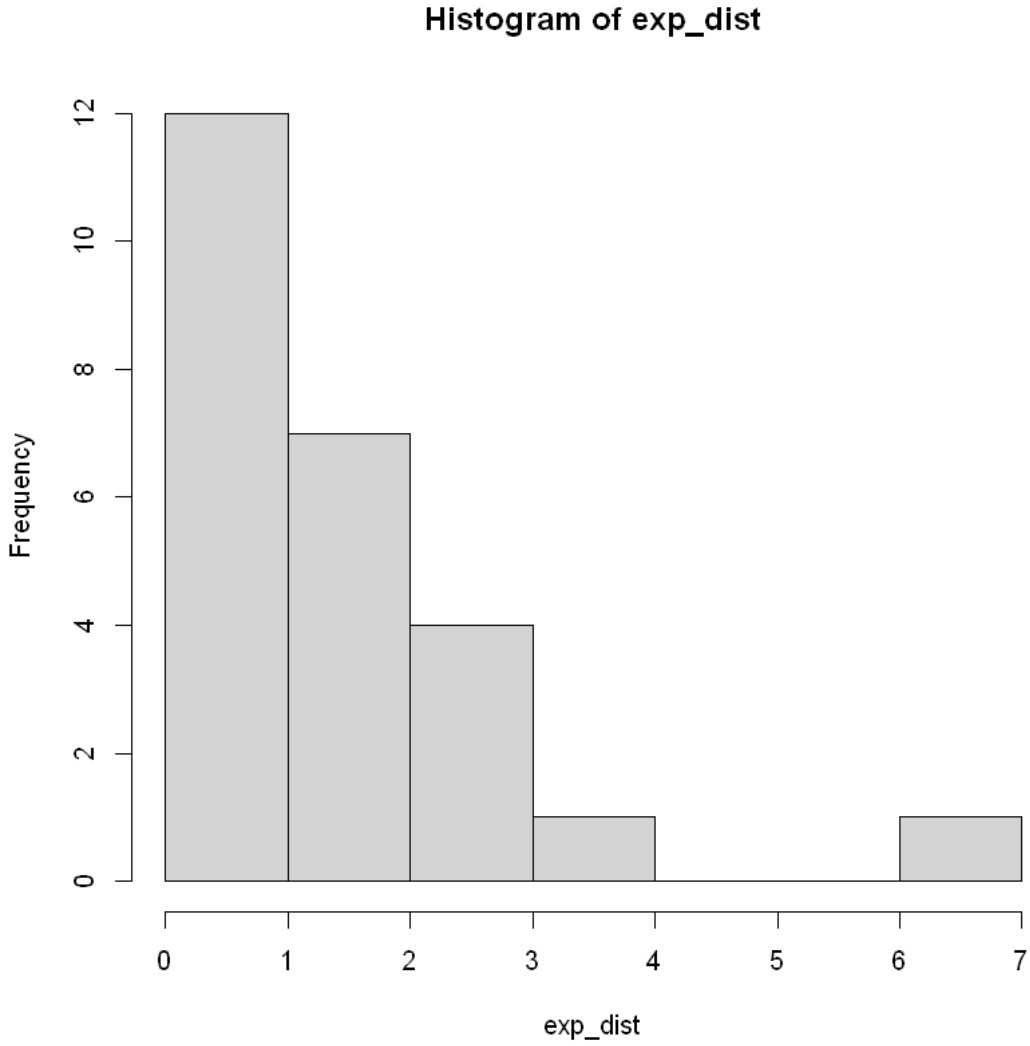

**Comment :** The points follow more or less the line, with points slightly above and slightly under the line, which means the sample quantiles are close to the theoretical quantiles and thus that the Hamilton score of women in group I follow a normal distribution.

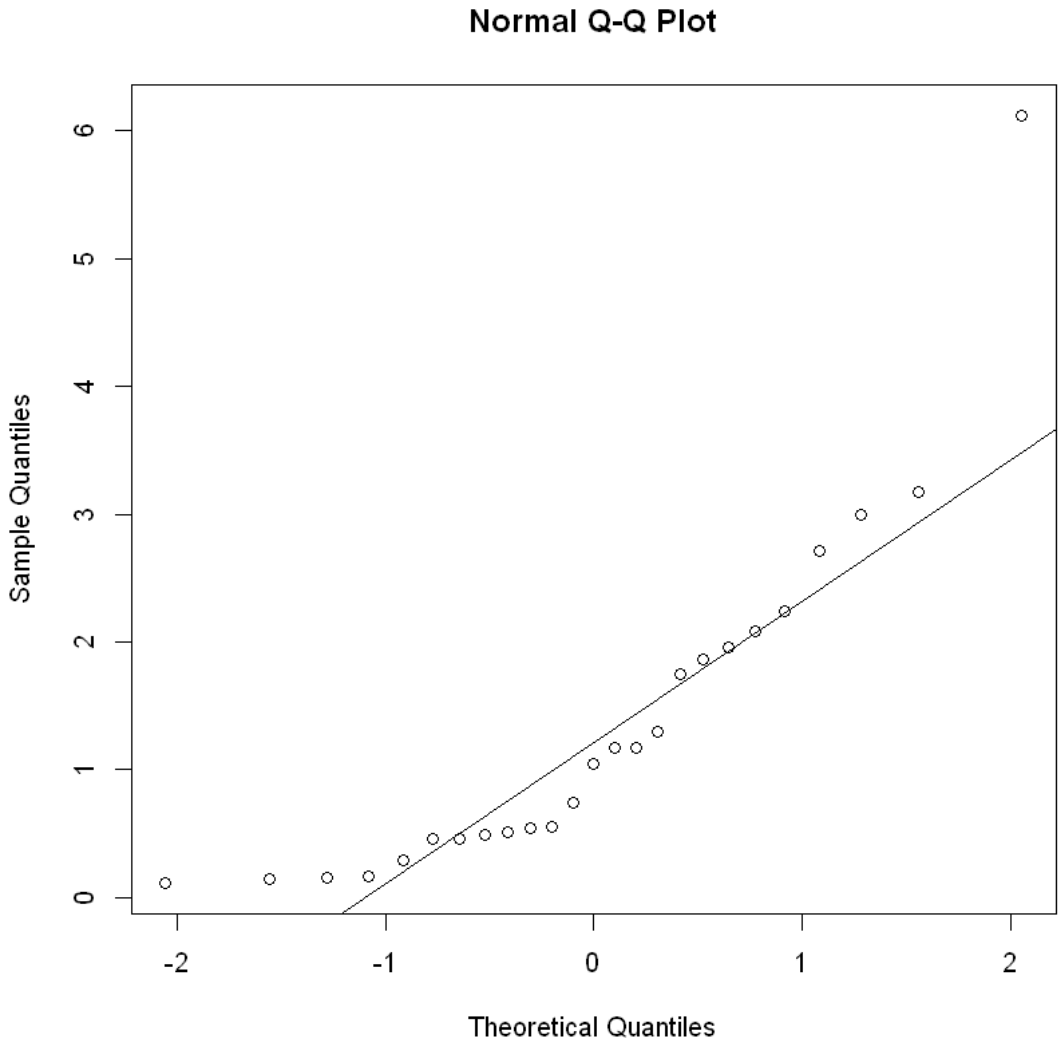
[6]:
```
# Additional example where a distribution is absolutely not normal

exp_dist = rexp(25)  # Random exponential distribution

hist(exp_dist)  # Display the distribution

qqnorm(exp_dist)
qqline(exp_dist)
```


Histogram of exp_dist
Frequency
0
2
4
6
8
10
12
0
1
2
3
4
5
6
7
exp_dist




**Statistical method: the Shapiro-Wilk test** It is also possible to use a statistical test approach, for example, by calculating the variable W of the Shapiro-Wilk test. calculating the variable W of the Shapiro-Wilk test. For this, the shapiro.test function both calculates this W value and compares it with its theoretical distribution under the null hypothesis of data normality.

H0 : The data follows a normal distribution

H1 : The data doesn't follow a normal distribution

You can use this test through the use of the function `shapiro.test()` function

[7]:
```
# Shapiro's test
shapiro.test(women_g1)
```

```
        Shapiro-Wilk normality test

data:  women_g1
W = 0.94764, p-value = 0.614
```

**Conclusion**:

We observe p = 0.614, which is higher than our $\alpha$ threshold of 0.05, thus we cannot reject H0, and as such, the Hamilton score of women follows a normal distribution.

### 1.3.4 3.4 Calculating the confidence interval

The mean $m$ calculated from our sample is an estimate of the mean $\mu$ Hamilton score for the female population. the mean $\mu$ of the Hamilton score for the female population. $m$ fluctuates as a function of the the sample we are analyzing. It is therefore a random variable with a normal distribution a normal distribution with expectation $E[m]$ and variance $V[m]$ if Hamilton's score itself is distributed itself according to a normal distribution with mean $\mu$ and variance $\sigma^2$.

For a sample of size $n$ and for a confidence interval at a given threshold $\alpha$ (often 95%) is calculated with the following formula:

$$CI = \mu \pm Z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

with $Z_{1-\frac{\alpha}{2}}$ the quantile of the Student's law.

There is no function in **R** that can be used to calculate a confidence interval directly. So to calculate the 95% confidence interval of the Hamilton score, we'll use the `CI()` function, which allows you to do this in a single line.

This function will take as arguments the data (x) and the chosen threshold.
For example, for a 96% confidence interval the function call should be: `CI(x, 0.96)` (4% $\alpha$ threshold).

**IMPORTANT NOTE: that you'll need to copy-paste this function at the top of each new session notebook - an example will be provided in third session notebook**.

[10]:
```r
# You don't have to understand the code below nor to edit it
# Simply to know how to use it and copy-paste it
C.I <- function(x, alpha){
    alpha <- 1-alpha
    mu = mean(x)
    sigma <- sd(x)
    n = length(x)
    CI_inf = mu - (qt(1 - alpha/2,n-1) * sigma/sqrt(n))
    CI_sup = mu + (qt(1 - alpha/2,n-1) * sigma/sqrt(n))
    CI = c(inf = CI_inf, sup = CI_sup)
    return(CI)
```

```
}
```

[14]:
```
# Default 95% confidence interval for mean Hamilton score
# of women in group 1
C.I(women_g1, 0.95)
```

**inf** 39.3428909153857 **sup** 43.9298363573416

The true mean lies between the bounds: $[41.570; 41.703]$

*Application: compute the 98% confidence interval for the Hamilton score of women in group I.*

[13]:
```
# 98% default confidence interval for mean Hamilton score
# of women in group 1
C.I(women_g1, 0.98)
```

**inf** 38.7915544568985 **sup** 44.4811728158288

What are the bounds of the confidence interval this time?

$[38.781; 44.482]$

Larger bounds because higher confidence threshold

## 1.4 4. Exercises

### 1.4.1 Exercise 1

1. Compute the confidence interval of the Hamilton score for men and then for women for group 1, checking the conditions of application (at the 95% threshold).
2. Compute the confidence interval of the Hamilton score for men, then for women, for group 2, verifying these conditions of application (at the 95% threshold).
3. What do you conclude?

[22]:
```
## 1.
# Extract data
men_g1 = depression[depression$sex_g1 == "H", "hamilton_g1"]

# Check sample size
length(men_g1)

# -> Sample size >30 so no need to check normal distribution
# Compute the confidence interval
C.I(men_g1, 0.95)

# For women of group I the conditions of application are checked above
C.I(women_g1, 0.95)

## 2.
men_g2 = depression[depression$sex_g2 == "H", "hamilton_g2"]
```

```r
# Sample size
length(men_g2)
# => sample size >30 so no need to check normal distribution
C.I(men_g2, 0.95)

women_g2 = depression[depression$sex_g2 == "F", "hamilton_g2"]
length(women_g2)
# => sample size <30 so need to check if normal distribution

## First method: Shapiro Test
# let's perform a shapiro-test
# Hypotheses:
# H0: the Hamilton scores of women in group II follows a normal distribution
# H1: They don't follow a normal distribution
shapiro.test(women_g2)

# We observe p = 0.83 >>> alpha = 0.05, so we can't reject H0, and we conclude␣
 ↪that the scores of women in group II follow a normal distribution

## Second method: quantile-quantile plot
qqnorm(women_g2)
qqline(women_g2)
# It seems that the points follow a normal distribution, though with only n =␣
 ↪5, it's difficult to have a clear conclusion anyway

# We can proceed to compute the confidence interval
C.I(women_g2, 0.95)
```

39

**inf** 36.3595508838125 **sup** 39.5891670649054

**inf** 39.3428909153857 **sup** 43.9298363573416

45

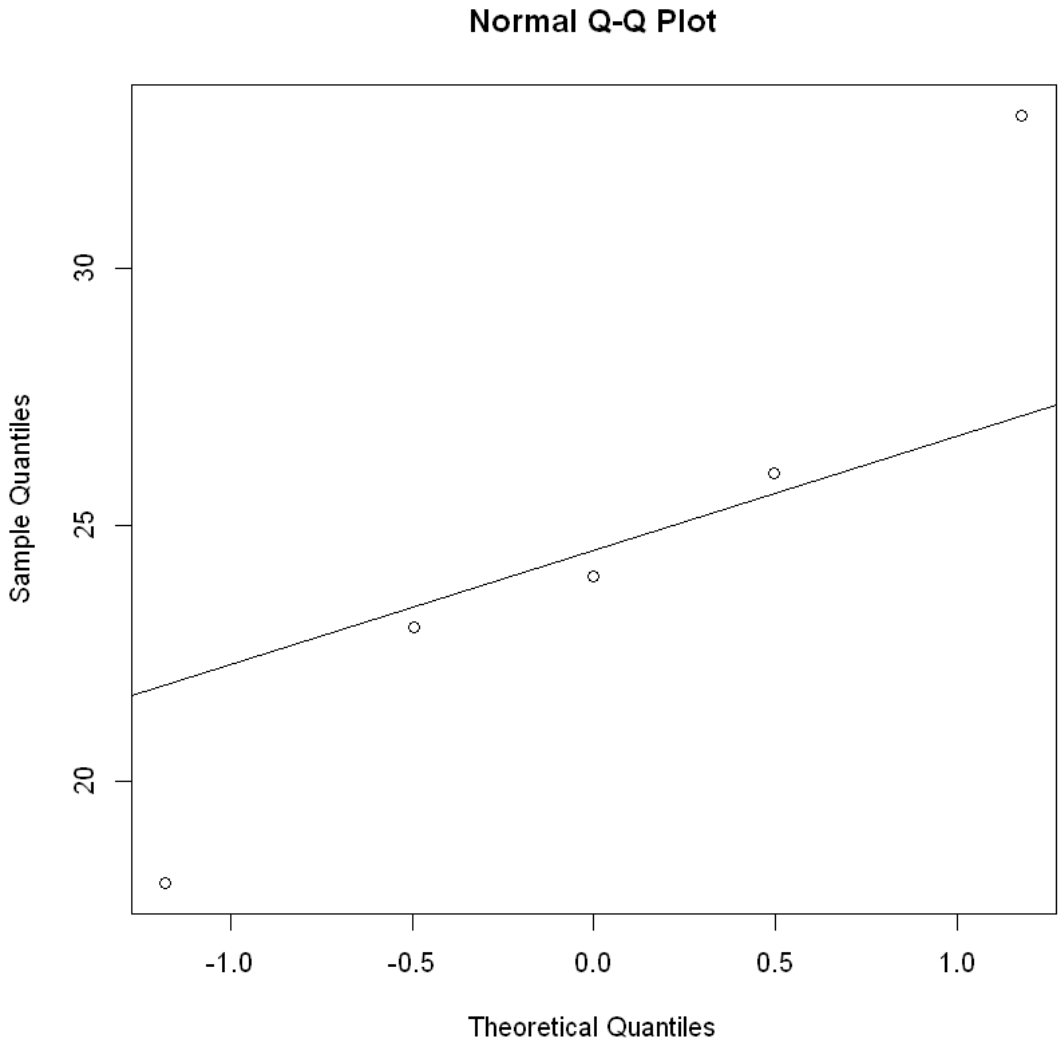
**inf** 23.0026458856954 **sup** 26.0195763365269

5

```
        Shapiro-Wilk normality test

data:  women_g2
W = 0.96258, p-value = 0.8258
```

**inf** 18.0332160011029 **sup** 31.5667839988971




We observe higher values in group I then in group II and for both a larger CI for women then men, suggesting that we know the average of men with more precision (we do have more observations)

### 1.4.2 Exercise 2

1. Extract a subset of the data, to be named U1, containing the Hamilton scores of group 1 in sparsely urbanized areas (i.e. `urbanization_g1=="A"`).
2. Calculate the confidence interval of U1, verifying the conditions of application (at the 95% threshold).
3. Extract a subset of the data, to be called U2, containing the Hamilton scores of group 1 in urbanized areas (i.e. `urbanization_g1=="B"`).
4. Calculate the confidence interval of U2, verifying the conditions of application (at the 95% threshold).

[31]:
```r
## 1.
u1 = depression[depression$urbanization_g1 == "A", "hamilton_g1"]


## 2.
length(u1)
# n < 30 -> let's check normality

qqnorm(u1)
qqline(u1)

# According to the plot seems almost normal though has non normal tails

C.I(u1, 0.95)

## 3.
u2 = depression[depression$urbanization_g1 == "B", "hamilton_g1"]

## 4.
length(u2)
# n < 30 -> check normality

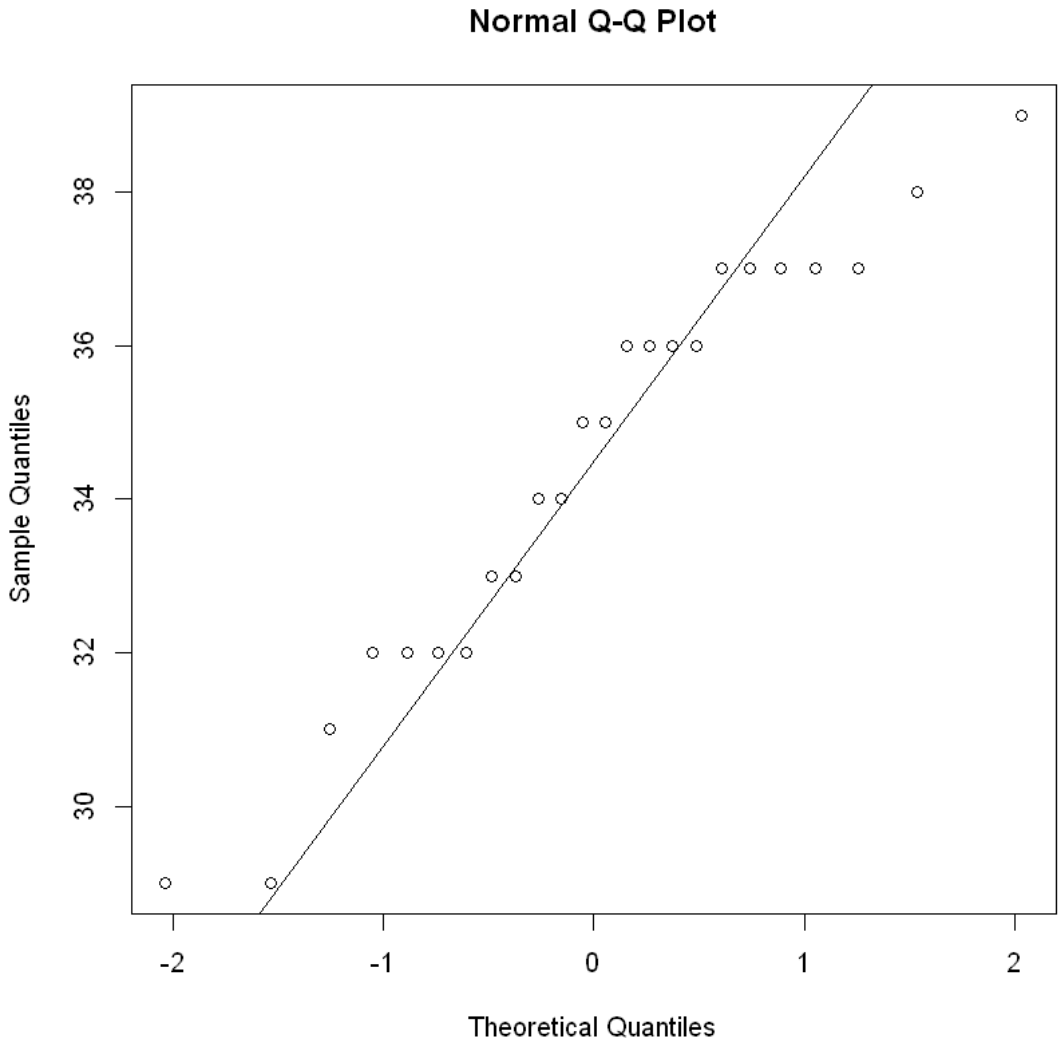
qqnorm(u2)
qqline(u2)

shapiro.test(u2)
# We observe p = 0.02 < alpha = 0.05, so we reject H0 and the scores don't␣
 ↪follow a normal distribution thus we can't compute the confidence interval
```

24

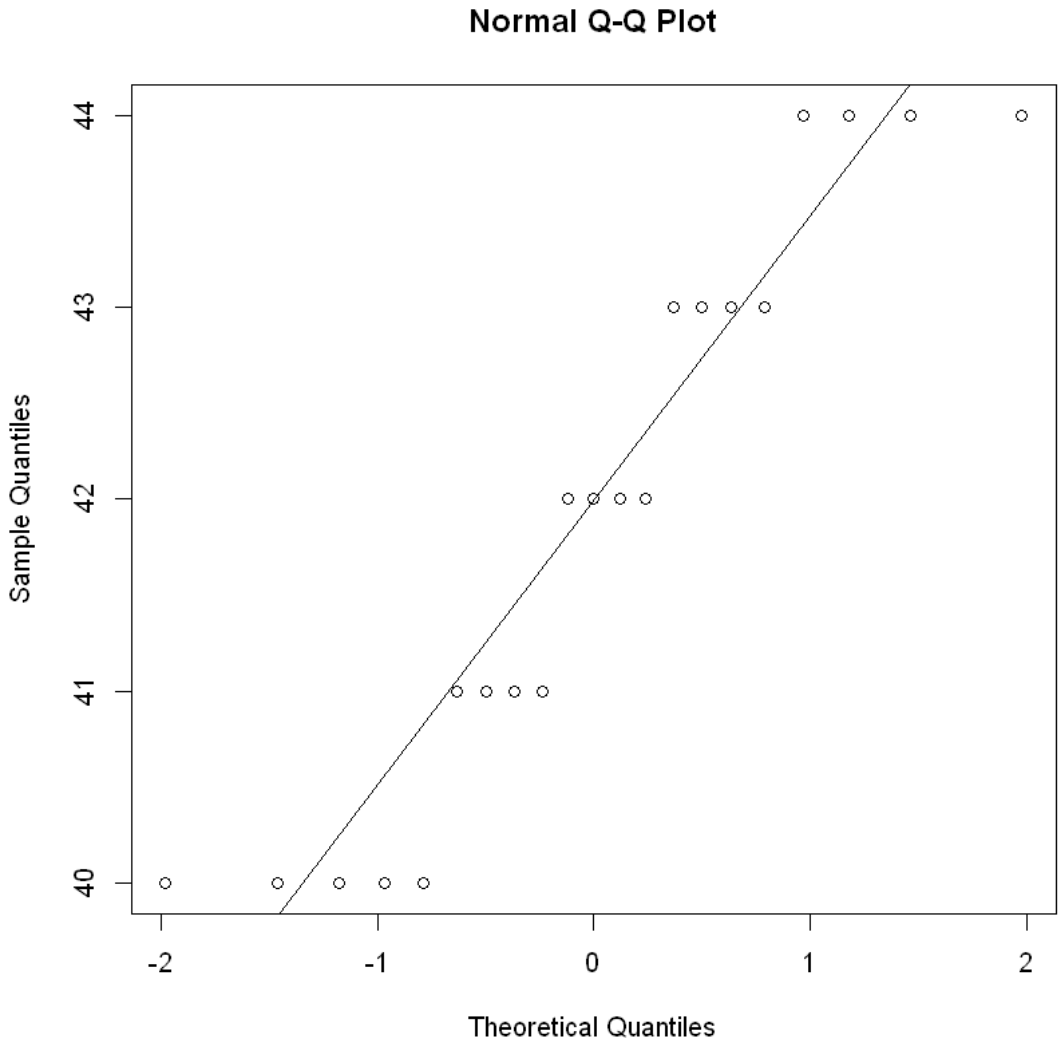
**inf** 33.2903840184366 **sup** 35.6262826482301

21




```
        Shapiro-Wilk normality test

data:  u2
W = 0.88686, p-value = 0.01962
```


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
40
41
42
43
44
-2
-1
0
1
2